

Sport-decision aids and the “CSI-effect”: Why cricket uses Hawk-Eye well and tennis uses it badly

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Abstract

Technologies of visualisation and measurement are changing the relationship between spectators and match officials at sporting events. Umpires and referees find themselves under increasing scrutiny and sports governing bodies are experimenting with new technologies and additional “off-field” officials in order to preserve the legitimacy of decision-making. In this paper, we examine how technologies are being used in a number of sports, paying particular attention to the way in which uncertainty and indeterminacy are conveyed to viewers and spectators. The contrast between cricket and tennis is particularly instructive in this respect as the same technology is used in two very different ways. The paper concludes with a series of recommendations for implementing sports measurement technologies whilst preserving the traditions of individual sports and enriching technological culture.

Keywords

innovation, interaction experts/publics, media and science, popularisation of science, public understanding of science, representations of science, studies of science and technology

1. Introduction

The experience of being a spectator at a major sporting event has changed. At live events, big screens provide live video feeds and replays of the on-field action for watching fans. At home or in public bars, viewers can get live coverage, commentary and analysis along with replays of key moments from a number of different camera angles. After the event, pundits and players debate contentious decisions using slow motion replays and computer graphics. As a result, referees and umpires are under unprecedented scrutiny with technology seen as both the source of the problem and its solution. It is, therefore, rather surprising that, despite this combination of cutting-edge technology, a mass audience and high-stakes decision-making, sports decision aids have not been a topic of research within the public understanding of science.¹

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Here we begin to fill the gap by comparing the representation of science in sports broadcasting with the representation of science in other popular media. We focus on two issues. First, how are different technologies used in different sports and secondly, to what extent does the coverage of sporting events that use these technologies reproduce the same “under-socialised” vision of science found in other media genres?

We begin by summarising some relevant existing literature. We then explain the terminology we use to classify different technologies and analyse decision-making in sport. The main empirical sections of the paper follow, describing how different technologies are used in a range of sports and highlighting how their limitations are either accommodated or ignored. The contrast between cricket and tennis is instructive in this respect. We conclude by reflecting on the ways these technologies could enrich public discourse about science and technology.

2. Existing work which bears upon the analysis

The Sociology of Technology (SCOT) is concerned with the way technologies can be interpreted (Pinch and Bijker, 1987; Bijker, 1995, 2010). Here we discuss the ways that sports decision aids (SDAs) such as video replays and Hawk-Eye are interpreted by different groups. In the spirit of early SCOT we could be said to be exploring how various SDAs mesh with the socio-technical frames of relevant social groups including governing bodies, players, referees, fans, spectators and technologists.

We are also interested in the implications of these differences for the public understanding of science. If different socio-cultural contexts give rise to different instantiations of the “same” technology they might also, as Bijker has speculated in relation to coastal defences, give rise to different levels of “technical literacy” (Bijker, 2007). Thus, as with recent developments in SCOT, our argument covers both the interpretive flexibility of artefacts and the normative aspects of their development and use. We claim that STS could be a participant in these debates and should work to encourage the stabilisation of technologies that make the assumptions that underpin their use more visible (e.g. Collins and Evans, 2008).

This also links our work to a broader tradition of public engagement activities that includes science education (e.g. the Nuffield Science curriculum projects of the 1960s) and public participation (e.g. Wilsdon and Willis, 2004; Haddow et al., 2008). In each case, the idea is that pupils and citizens benefit from the opportunity to experience science as a social practice rather than a set of facts to be learnt. Whilst there are legitimate questions about how far specific attempts to enact these principles have succeeded (e.g. Atkinson and Delamont, 1976 for education; Irwin, 2001 for public participation), we are primarily interested in the principle, namely that the uncertainties and choices of science should be made visible (see e.g. Driver et al., 2000; House of Lords, 2000).

There are some reasons to believe that making these uncertainties visible in broadcast coverage should have some effect. After all, there is a substantial literature about the portrayal of science in the media that assumes some influence or connection (e.g. Nelkin, 1995; Miller et al., 1998). At the time of writing, one of the most pervasive images of science-being-used is what Michael calls the “technoscientific police procedural” (2006: 90). In this very popular genre, forensic science is used to solve otherwise baffling crimes. In the case of the *CSI* franchise, which is probably the most well-known of the shows (*Cold Case* and *NCIS* are others), Michael describes the typical episode as follows:

The *CSI* shows unfold as a dramatic, but patient, accumulation of evidence ... laboratories are pristine, equipment works without breaking down, and the visualization of evidence, enhanced as it is by

computer graphics, reveals the most obscure and intricate physical and physiological processes ... the refrain, common to all characters in those moments when the plot demands that they play the rational foil to an irrational or distracted colleague, [is] of letting the evidence “speak” for itself. (Michael, 2006: 106)

In other words, *CSI*-science conforms closely to the canonical model, in which “the truth will out” as the scientific method erases all social and technical contingencies (Mulkay and Gilbert, 1984; Collins, 1992). It is this asocial representation of science that underpins concerns about the so-called “CSI-effect” on the public understanding of genomic sciences. As Ley et al. write, the show’s

... dominant messages about forensic DNA testing – specifically, that it is easy, quick, routine and epistemologically very strong – [echo] broader media and cultural discourses that glamorize and valorize genetic knowledge, science and technology. (Ley et al., 2010: 13)

This, it is suggested, encourages viewers to develop an unrealistic faith in the utility, accuracy and power of genetic science.²

There is, however, an important difference between *CSI* and sports broadcasting. Although both are entertainment genres, sport is not fiction. Like news broadcasting it claims to show events as they really happened. This provides a further reason for taking the representation of science in sports broadcasting seriously. In a study explicitly designed to explore the relationship between the reporting of science-based news and the public understanding of these topics, Hargreaves, Lewis and Speers (2003) found:

The news media clearly play a role in informing the way people understand science. Our study suggests that most people are aware of the main *themes* or *frameworks* of media coverage of science related stories. Information that is subsidiary to these themes, be it part of the background to a story or information that does not recur (such as the passing of legislation) is unlikely to get across. (p. 52, emphasis in original)

To give the most striking example from the three stories they examined, only about a quarter of respondents in a representative survey knew there was very little evidence to support Andrew Wakefield’s claim that the MMR vaccine could cause autism. Instead, about half (53 per cent) believed that the scientific evidence was evenly balanced and the remainder (20 per cent) believed the evidence supported Wakefield. Hargreaves et al. attribute this outcome to the journalistic convention of balance and to Prime Minister Tony Blair’s refusal to confirm that his own child had received the vaccine.

Of course, showing that news coverage involves some form of selection and framing (see e.g. Marks et al., 2007; Stewart et al., 2009), or that audiences approach media reports with some scepticism (see e.g. Philo, 1999; Miller et al., 2008), is not new. Rather, our concern is that, as with the fictional representation of science, these topics are all the subject of substantial social science research. In contrast, sports broadcasting, which forms a significant part of the public culture that individuals engage with and which uses an increasingly complex array of technologies, has yet to be subject to similar scrutiny. Our aim in this paper is to begin to redress this balance and examine the ways in which video replays and other SDAs are used in sports broadcasting. Specifically, do they, like *CSI*, reinforce popular discourses about science and technology or do they allow viewers to gain some understanding of the complex chains of inference involved in drawing conclusions from remote sensing and digital imaging technologies?

3. New terminology

To analyse the way sports decision aids and broadcast technologies enable different kinds of viewer experience some new technical terms are needed. These are explained in detail elsewhere (Collins, 2010) so we here define them only briefly:

- *Ontological authority* is the power granted to match officials (or machines) to create reality (e.g. whether the ball was “in” or “out”).
- *Epistemological privilege* is the advantage someone or something has when it comes to knowing what happened (e.g. due to a special vantage point or skill). One consequence of video replays has been to transfer *epistemological privilege* from on-field officials to spectators.
- *Transparent justice* means that the correct decision can clearly be seen to be made (i.e. that justice is seen to be done). Where technologies like video replays support the on-field official they enable *transparent justice* and legitimate *ontological authority*.
- *Transparent injustice* is the complement of transparent justice when justice is clearly seen not to be done. Where video replays or other technologies show a mistake has been made they reveal *transparent injustice* and undermine the *ontological authority* of on-field officials.
- *Presumptive justice* is the assumption, when you cannot see clearly what is going on, that you would make the same decisions if you were the match official, or that you could do no better. *Presumptive justice* is the default position and prevails where SDAs are not used. Where SDAs are used, *presumptive justice* is invoked whenever the technology is unable to provide clear evidence of a mistake.
- *False transparency* is the impression that justice is being seen to be done when it is not. In sport, this typically occurs when technologies are used without appropriate acknowledgment of their limitations.³

4. Technology and sport

We now examine how different sports use technologies, distinguishing between technologies that are used to *enhance perception* and those that are used to *take decisions*. Our starting point is the assumption that players, spectators and match officials would all prefer to see fewer egregiously bad decisions. As evidence for this, we would cite the vociferous complaints of players and spectators when instances of transparent injustice occur and the attempts in virtually all sports to improve decision-making, either by using new technologies or by increasing the number of officials. We should, however, stress that in what follows we are primarily concerned with the ways in which these technologies are presented. A detailed study of their reception and interpretation would require a very different kind of research project to the one reported here.

Using technology to enhance perception

The first set of examples is concerned with the ways in which television and similar technologies are used to enhance perception and support the decision-making of match officials. The main tool here is the “slow motion replay,” though the “photo finish” is another example. Where these technologies are used, the replays are usually reviewed by a newly created class of “off-field” officials who then advise the on-field officials. Because only ubiquitous skills are needed to interpret these

images (e.g. did the ball cross the line), the epistemological privilege of the off-field officials is matched by that of the spectators, assuming they understand the relevant rules.⁴ It is this parity of vantage point that allows the off-field officials to correct big mistakes and protect the ontological authority of the on-field officials.

Whilst there is an interesting study to be done about why particular sports, notably soccer, remain non-users of a technology, in this paper we are concerned with what happens when they are used. We begin by examining two incidents that took place during the test cricket series involving England and South Africa played between December 2009 and January 2010. The first occurred during the third match of the five match series and highlights an unproblematic use of technology to enhance perception. The second, which was far more controversial, exposes the limits of video technology and shows why some (fallible) human judgement remains an integral part of all decision-making processes.

The first incident took place just after the start of South Africa's second innings on the third day of the match. The South African opening batsman, Ashwell Prince, was given out by the on-field umpire, Daryl Harper, who appeared to think that Prince had started to walk off the pitch, thus confirming England's appeal for a catch. Prince asked for the decision to be reviewed. The video replay showed the ball had missed the bat by a considerable distance and actually hit Prince's leg, meaning that he should not have been given out. The following extract, from *The Guardian's* online coverage, gives an indication of how the decision was seen at the time:

How odd! England appeal for a catch behind down the leg side. Prince turned away from the wicket to compose himself, and umpire Harper seemed to think he was walking off. He wasn't. So after an implausible delay, Harper raises his finger [i.e. indicates that Prince should be given out]. Prince refers it, and it becomes clear that it was an absolutely awful decision. Really shocking. His bat was a full six inches away from the ball, which hit the back of his shin. So the **decision is over-turned** and Prince continues. What a bizarre piece of play.⁵

The second case also involves umpire Harper, though now in the role of off-field umpire, and occurred on the first day of the Fourth Test. In this case, the referral concerns England's opening batsman, Alastair Cook, who had been given out "leg before wicket" (lbw) by the on-field umpire. Cook asked for the decision to be reviewed and the subsequent controversy hinged on whether the ball that got him out was a legal delivery and, more specifically, whether or not the bowler's foot was behind the relevant line (the "popping crease") when the ball was delivered.⁶

For some viewers it was clear that the television replays showed the delivery was not legal and that the original decision should have been overturned. A sense of the certainty with which these views were held can be gleaned from the BBC's online coverage, which described the incident as follows:

[09:19 am] Full and fast next – bang into the front pad, huge appeal – GONE! Wait, though – they'll refer, this looks like a no-ball – Morkel [the bowler] has failed to ground his heel behind the line. So why are they still showing replays of the ball hitting the pad? Don't tell me that third umpire Daryl Harper hasn't spotted that ... He's given it out!

[then, 8 minutes later, at 09:27 am]

You couldn't make it up – certain no-ball, and Harper has managed to miss something that he's just been shown four times on [*sic*] slow-motion on a big screen right in front of his face⁷

In contrast, *The Guardian's* online coverage reported the same incident as follows:

Cook is given out LBW [...] and decides to refer it. Unfortunately for everyone, the man in the box is umpire Harper. Hawkeye says it pitched in line and was going on to hit the stumps ... but ... England are furious anyway. The replays really make the delivery look like a no ball. Morkel's foot was ever-so-close to being over the line as he let go of the ball. There was possibly just the slightest sliver of boot behind the line. Anyway, Harper isn't going to over-turn the decision unless he has to, so the bowler gets the benefit of the doubt. The decision is upheld

[then, a short time later]

To clarify: I don't agree with Nasser Hussain [ex-England captain and commentator for Sky Sports] about Harper's decision on that referral: there was just enough boot behind the line, the slightest suggestion of leather.⁸

Unlike the first case, here there is no clear and obvious mistake. Instead, there is a borderline judgement about where the bowler's foot was at a crucial split-second. The difficulty is that the decision now takes place at the limits of the camera's resolution and the data are too ambiguous to be easily classified using a scale that allows only two positions.⁹ In these circumstances, the ontological authority of the third umpire is needed to bring about a resolution to the debate and, in this case, the decision was upheld. Whilst some spectators clearly disagreed with this decision, others did not. The sociological point is that, because the ambiguity of the images is apparent to viewers, so is the difficulty of the decision. If, however, it is accepted that the role of technology is to help umpires avoid big mistakes then whether Cook is given out or the original decision is overturned does not matter. The dispute over how to interpret the video evidence demonstrates that no big mistake has been made and that a human judgement was needed in order to determine how the match should proceed.

To summarise this section, the adoption of video replay technologies stabilises a new socio-technical ensemble in which the epistemological privilege of spectators is increased (cf. Bijker, 1995). Indeed, it could be said that where video replays are used, on-field officials find themselves at the centre of an "inverse panopticon" where everyone can scrutinise the actions of the single person at the centre. Where these technologies are also used by off-field officials, this imbalance is corrected and the epistemological privilege of match officials is raised to match that of the spectators. Our main claim however, is that, because the marginal nature of close decisions is made visible by the technology, the system itself reveals the necessity of human intervention. In other words, there is no problem of public "misunderstanding" of science here as the uncertainty of the technology is already apparent in the quality of its outputs.

Using technology to take decisions

There is a class of technologies that do more than replay events for others to judge. These technologies process data to produce a simulation that is meant to represent what "really happened" and, using this output, can replace human officials for some decisions. One particularly well-known version of this kind of technology is the Hawk-Eye system, which is used in both tennis and cricket to determine the ball's bounce point and trajectory. As with video replays, our concerns relate to the ways in which these technologies are used and the extent to which any uncertainty or ambiguity is made visible. We begin our analysis by briefly setting out how these technologies work.

“Hawk-Eye” is an example of what we call a “Reconstructed Track Device” or RTD. RTDs use visible-light television cameras to follow the path of the ball and a procedure to filter the pixels in each frame. Certain pixels are taken to represent the position of the ball and others to indicate the position of the line or other features of the playing arena. The space and time coordinates of these pixels are represented numerically and a statistical algorithm reconstructs the flight and impact point of the ball and crucial features of the playing area by combining information about the pixels in the different frames with information about the size of the ball, the physics of its distortion, the width of the line, and so forth. From these calculations, the system then determines what decision should be given – for example, should the ball be called “in” or “out”?

In the case of Hawk-Eye, we also know that, when the system is used in cricket matches, it is operated by three people. Hawk-Eye Innovations describes their roles as follows:

1. Lining up and calibrating the cameras
2. Measuring the pitch and the stumps which do vary from ground to ground
3. 1 member of staff is responsible for the virtual reality graphics and offers LBW replays and all the other Hawk-Eye features to the TV director.
4. The other 2 members of staff both are responsible for the tracking. They work independently of each other to provide redundancy, but are able to see a comparison of the two tracks. If they are different for any reason, they can be pro-active in working out why rather than being re-active after a LBW appeal. On a ball by ball basis they would do the following:
 - a. hit a button to tell the system that a ball has been bowled and trigger the tracking
 - b. *manually fine tune the point on the trajectory where interception with the batsman was made. Automatically the system is only able to determine the interception point to the nearest frame of Hawk-Eye video running at 106 frames per second. This can be improved manually and is the only way to ensure that the interception point is accurate to 5mm.*
 - c. Tune settings to account for varying light conditions
 - d. Tune settings to deal with camera wobble¹⁰

What viewers see of this effort is the virtual reality graphic. This is either an image or a short video clip showing the path and impact point of the simulated ball. In the case of tennis, information about the likely distortion of the ball upon bouncing is used to estimate the size and shape of the contact footprint and the visually reconstructed bounce point is elongated to represent the way the ball compresses and skids as it hits the ground. These details, together with life-like colours and other details, give the simulation an appearance approximating the real setting, albeit with sharpened edges and idealised precision.

It is here that the potential for RTDs to contribute to a public misunderstanding of science is realised.¹¹ Because of its statistical nature, any RTD technology is subject to statistical uncertainty. However sharp the reconstructed images appear, they represent an “estimate” which has errors that will affect the accuracy of the estimated bounce point, the shape of the reconstructed footprint and the reconstruction of the line. The key question for both those concerned with the legitimacy of decision-making (i.e. the relationship between epistemological privilege and ontological authority) and the public understanding of science is, therefore, the extent to which these errors and their significance is made visible.

In an earlier paper (Collins and Evans, 2008) we discussed the accuracy of Hawk-Eye in some detail. Here we simply note the few facts needed to contextualise our current argument:

1. That RTDs have an error distribution is not denied by manufacturers. Hawk-Eye Innovations' website states that, based on testing carried out with the International Tennis Federation (ITF) in 2006, Hawk-Eye has an average accuracy of 3.6 mm in tennis.¹² More recent testing, this time in 2008 with the International Cricket Council (ICC), gives an average error of 2.6 mm for the pitching point in cricket.¹³ We do not dispute these figures or that they represent a remarkable technological achievement. Our point is simply that any average implies that there are larger errors as well as smaller ones. For example the following "triangular" distribution of 132 errors has an average accuracy of 3.3 mm.¹⁴

Size of error (mm)	0	1	2	3	4	5	6	7	8	9	10
Number of errors	22	20	18	16	14	12	10	8	6	4	2

2. It is believed by some that no projection or estimation is involved in the case of tennis.¹⁵ This is incorrect as all cameras have a finite frame-speed. If the frame-speed is, say, 100 frames per second, and the ball is moving at about 100 mph it will travel about 1.5 feet between frames; more if the ball is travelling faster.¹⁶ The difference between cricket and tennis is, then, a matter of degree.
3. The size of any error is likely to increase with the length of the track that needs to be predicted. This is particularly important in cricket where the application of the lbw rule involves projecting the flight of the ball after it has hit the pad in order to determine whether it would hit the stumps.

5. Uncertainty, epistemological privilege and ontological authority

To return to our argument, we have characterised the problem of using technologies to support decision-making in sport in terms of the harmony between epistemological privilege and ontological authority. When these work together to enhance perception spectators and officials share the same epistemological privilege. Marginal decisions will, of course, remain so some element of controversy will persist when spectators refuse to agree that a difficult decision was fairly taken (transparent justice) or that no-one could have done any better than the on-field official (presumptive justice). In the case of RTDs, however, something different happens. Now the technology can either provide information for officials to use in their decision-making (i.e. enhance epistemological privilege) or it can replace those officials altogether (i.e. assume ontological authority). We now show the significance of this choice by examining how the same RTD technology is used in two different sports.

Hawk-Eye in cricket

We begin with cricket's "lbw" decision, which effectively asks the umpire to decide whether or not the ball would have gone on to hit the stumps if it had not hit the batsman's pad.¹⁷ The decision is notoriously difficult as the umpire has to extrapolate from what did happen to what would have happened if the ball's flight had continued uninterrupted. Where the on-field umpire cannot be sure that the ball would have gone on to hit the stumps, the convention in cricket is that the batsman gets the benefit of the doubt and is given "not out." We now explain how this convention, which is an

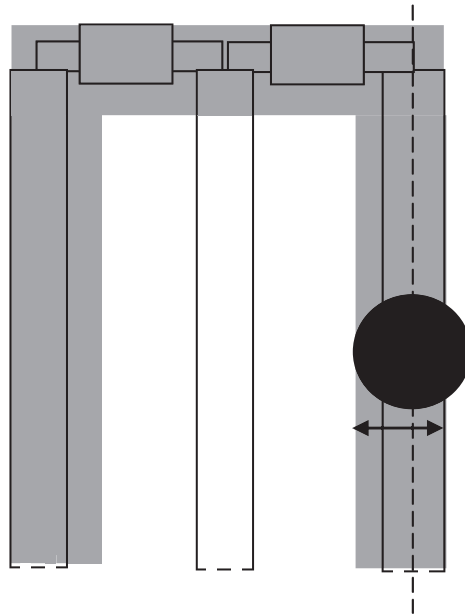


Figure 1. The grey box illustrates the zone of uncertainty on one edge of the wicket equal to half a stump and half a ball (not to scale).¹⁸

integral part of cricket's culture and ethos, is encoded into the rules governing the use of Hawk-Eye in the review of these decisions.

In the review system, where Hawk-Eye is used to check the umpire's decision, the crucial element is what we are going to call "the zone of uncertainty." This is an area about 55 millimetres (half the width of the ball plus half the width of one stump) around the edge of the stumps (see Figure 1).¹⁹ In this area, the RTD cannot over-rule the on-field umpire. Thus, in the case where an umpire gives a "not out" decision (i.e. decides that the ball would not have hit the wicket) and there is an appeal, the umpire is not over-ruled unless the *inside* edge of the ball is shown to be striking a point *c.* 55 mm inside the outer edge of the wicket. The ICC's guidelines express this as follows, saying that, when the inside edge of the ball is shown inside what we have called "the zone of uncertainty":

the third [off-field] umpire should report that he does not have a high degree of confidence so the original on-field decision remains.²⁰

Similarly, if the umpire gives an "out" decision, this is only overturned if the RTD suggests that no part of the ball was inside the zone of uncertainty.

There are also zones of uncertainty in respect of the other two judgements that have to be made to apply the lbw rule correctly. These concern the impact point of the ball on the ground as it bounces before striking the batsman and the impact point of the ball on the pad. These also have similar rules that determine when the RTD can be used to over-rule the on-field umpire and when the original decision should stand. A further feature recognising the fact that an RTD cannot entirely replace human judgement is the following rule used by the ICC:

if the point of impact [on the pad] is greater than 250cm from the stumps, the third umpire will inform his on-field colleague of the exact distance, the approximate distance from the point of pitching to the point of impact and where the ball is predicted to hit the stumps. The on-field umpire will then apply the normal cricketing principles concerning levels of certainty in making his final decision.²¹

In our language, the zones of uncertainty accomplish two tasks. First, they take explicit account of the uncertainty associated with the RTD technology and convey that to viewers. This is important because the virtual reality graphics of the RTD are not subject to the same limitations of resolution and viewpoint as conventional video replays where the uncertainty is conveyed by the quality of the image. Secondly, the zone of uncertainty, together with the 2.5 m rule, defines what counts as a big mistake. If the ball is shown to be hitting the wicket within this zone there is no big mistake and the on-field umpire retains epistemological privilege (e.g. by virtue of experience, viewpoint or training). In contrast, where Hawk-Eye shows the ball passing clearly within or completely outside the zone of uncertainty, the on-field umpire is over-ruled and the original decision overturned.²² From the perspective of STS, this way of using RTDs seems highly commendable as it conveys the uncertainty in a clear way whilst also allowing the technology to contribute to improved decision-making.

Hawk-Eye in tennis

In tennis, at least as it is being played at the time this paper is being written, the use of RTDs such as Hawk-Eye is very different. In tennis, there is no zone of uncertainty and hence no recognition that there is any error associated with the technology. Instead, the RTD makes its decision based on the modelled reconstruction and gives a binary “in” or “out” decision with no reference to any potential error or uncertainty. Here, in contrast to cricket, we have a *CSI*-style of presentation in which ontological authority is granted to the technology and the viewer encouraged to believe that the clarity of the reconstructed image gives them the epistemological privilege.²³ This is what we call false transparency and, to the extent that it conceals the nature of the decision that is being made from spectators it appears to be wrong.

The solution for tennis in the case of Reconstructed Track Devices

From an STS perspective, concerned with the public representation of technology, it seems clear that, like the ICC, the ITF should adopt rules that acknowledge the uncertainty inherent in the technology and make the operation of these rules as clear and transparent as possible. Here we are being normative and so must address head on the counter-argument that is made by those who argue that using RTDs for close calls is to be preferred as it is the most accurate system for determining where the ball has bounced.²⁴

First, we would argue that defining the problem in terms of accuracy is part of the problem. Instead, we argue that a better way to frame the problem is as one of eliminating *avoidable* errors and, to do this, millimetre accuracy is not needed. Indeed, makers of other sports decision aids have told us that it is a mistake to promote SDAs on the basis of accuracy alone as it is impossible to eliminate all errors. Instead, these manufacturers see the technology as solving a different problem, namely avoiding prolonged disputes between players and umpires. We, in turn, see it as a question of bringing epistemological privilege back into harmony with ontological authority. In either case, if the goal is to avoid big mistakes, then simply showing that the ball bounced very close to the line is enough to show that, as in cricket, the umpire’s decision was reasonable.

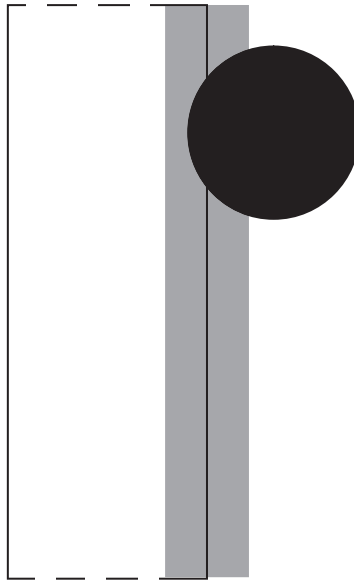


Figure 2. Zone of uncertainty on the crucial edge of a tennis line: if the trailing edge of the ball is shown to be in the zone the umpire's initial decision stands (not to scale).

Secondly, granting ontological authority to a technology, even when done in the name of reducing error, changes the game. Under the ITF's current system, there is an incentive for players to appeal against close decisions as there is always a statistical chance that the technology will give the call in their favour. If, however, the RTDs could only over-rule match officials when a big error was detected this would eliminate transparent injustice but also discourage players from "gaming" the system by appealing against crucial decisions irrespective of whether they really believe the ball was miscalled.

Given this, there seems no strong argument against the proposal that, like the ICC, the ITF should introduce zones of uncertainty. Once more, if the trailing edge of the ball entered the zone of uncertainty the umpire's original line call would be the default (see Figure 2). This would return ontological authority to the umpire except where a large mistake – that is where the RTD shows the trailing edge of the ball to be outside the zone of uncertainty – was seen to be made.²⁵

How wide should the zone of uncertainty be? A reasonable minimum would be that used in the ITF's own test procedures: 5 mm either side of the crucial edge of the line. To take account of less than optimum conditions, the "tails" of the statistical distribution and the blurred edges of the lines themselves, it might be wise to make the zone wider, though by how much is not a matter for STS.²⁶ Instead, our role is to explore the consequences of the current emphasis on accuracy and here we can be clear – the uncertainty inherent in RTDs should be made visible. Better rules for using RTDs are part of this but transparency could be further enhanced by the re-introduction of television replays for calls outside the zone of uncertainty. These would further reassure viewers that justice was being done in these cases and put an end to the embarrassment of having both players and umpire agree that the ball was miscalled by the RTD, as has happened, but still having to accept its decision.²⁷

6. The social construction of sports decisions aids

So far we have shown that, where SDAs are used, they form part of a socio-technical ensemble linking spectators, governing bodies and broadcasting organisations. We have also argued that SDAs should be used to assist human umpires but that, in marginal decisions, it is the umpires who should retain the ontological authority.

That said, we do not deny that viewers, umpires and governing bodies all have a legitimate interest in seeing uncertainty reduced and big mistakes eliminated. As noted above, however, uncertainty can never be eliminated completely and all technologies will encounter some borderline cases where human judgement is needed. The normative claim that follows from the STS literature on this point is that this uncertainty should be made visible to viewers and spectators.

We have also shown that the “problem” the technology is used to solve can be framed in different ways. In this final section we develop this idea further and suggest some alternative criteria that might be used to evaluate SDAs. In doing so, we follow constructivist accounts of technology assessment (e.g. Rip et al., 1995) to argue that it would be desirable, both in terms of public understanding of science and in terms of democratising technologies, if the available options were debated more openly. For example, it might be the case that accuracy is not the only dimension on which such systems might be judged. Other criteria might be accessibility for lower levels, skills needed to interpret outputs, or the potential for data generated to be used for other purposes.

We now conclude our discussion by describing some existing technologies and suggesting how they might rate on these different dimensions:

- **“Tennis Electronic Lines” or TEL.** TEL bases its estimate of ball bounce point on the interaction of a metallised ball with wires embedded in the court’s surface. It is the only device for which we have been shown an error distribution (see note 14) and, if this represents its performance in all conditions over long runs, it would appear to meet the accuracy criteria. Compared to the other systems TEL has the great advantage that it can be operated by the players and does not need specialist operators or other attendants. It is, therefore, the only device that could be installed in local tennis clubs and sports halls and be used at all levels of the game.
- **“Hot Spot.”** “Hot Spot” uses infrared cameras to image the heat signature caused by the impact of the ball with the ground. It has not yet been used in tennis but is used in cricket to indicate whether a ball has touched the bat or the glove of a batsman before being caught. Like video replays, Hot Spot will not provide clear evidence in all cases. In addition to low intensity impacts that generate little friction it is also likely that the clarity of its output will be affected by ambient conditions. For example, the image should be clearest when conditions are such as to allow the heat images of the line, the impact point and the ground to be clearly different. Other factors that may also explain the relatively low take-up of Hot Spot, especially outside cricket, are the high costs of the cameras, the need for specialist operators and the limited scope for re-using the data generated.
- **Auto-Ref.** Auto-Ref is an RTD like Hawk-Eye and requires specialist technicians and operators. We have been told that it uses 12 high-frame-rate colour cameras (usually running at about 220 frames per second but capable of much higher speeds) and that this makes it more accurate than the rough estimates presented in this paper. Its manufacturers also claim that it can deliver decisions in ‘real time’ and is less likely to be confounded when the path of a ball is superimposed on a player or a line than systems (if these are still in

use) that employ a smaller number of black-and-white cameras. Nevertheless, some zone of uncertainty will remain and should be acknowledged in its use and output. As an RTD, Auto-Ref will appeal most to those for whom the cost of collecting the data can be offset against the benefits gained from enhanced replays and performance analysis.

We must make it clear that this list simply sketches out the possibilities. The evidence upon which it is based is thin and uneven. We have obtained considerable help from the makers of TEL and Auto-Ref (e.g. see note 14 and the Acknowledgement), very little from the makers of Hot Spot and virtually nothing from Hawk-Eye that was not already in the public domain. Even from those from whom we have obtained most help, we can only report what they tell us. We have tried to add what we can deduce from first principles about the nature of the different technologies and, on this basis, to set out some of the different dimensions upon which they might be evaluated. To return to public understanding, we believe that nothing but good could come from a more open and public discussion of what these technologies can actually deliver. We also believe that at least some makers of these devices would welcome such a discussion.

7. Summary and conclusions

The introduction of new technologies has created new challenges for umpires and referees. These have come about because epistemological privilege has been re-distributed, leading to disharmony with ontological authority and to transparent injustice or, with some devices, false transparency. We have shown that many of these problems could be easily and cheaply resolved if the purpose of the technology was no longer seen as getting things exactly right but as avoiding big mistakes.

The difference between these two approaches was illustrated by the differences between cricket and tennis. Cricket, we argued, has implemented technology in a way that makes its uncertainties clear to all. In so doing, it also contributes to enriching public discourse about technology by making scientific issues of measurement error a visible part of its decision-making. In contrast, tennis uses technology in a way that resembles the genetic science of *CSI* crime fiction. We have argued that tennis would be better served, as would the viewing public, if the ITF adopted a similar procedure to that used in cricket.

To put these arguments in more general terms, our central claim is that SDAs should be used to ensure that the large, visible, errors that lead to transparent injustice are eliminated while the normal traditions of judgement based on human perception and presumptive justice are maintained in the case of marginal decisions. This would also maximise the continuity with the traditions of the individual sport and its practice outside professional or other high profile competitions.²⁸

In practical terms, these principles can be codified as four simple recommendations:

1. The legitimacy of on-field officials depends on their ontological authority which, in turn, derives from their epistemological privilege being at least equal to that of the spectators. All changes to on-field decision-making should, therefore, be taken with this principle in mind.
2. Where new procedures or technology are unable to identify a clear mistake, the social conventions of the sport should prevail and the original decision of the on-field official upheld.
3. In these marginal cases, some explanation of the decision that has been taken should be provided for spectators. In the case of RTDs this is provided by the zone of uncertainty.

In other cases, such as video replays, the discussions between the on-field officials and the off-field officials should be relayed to spectators and viewers.

4. There are many other criteria that could also be used to evaluate different SDAs and their adoption should include some explicit discussion of how these are affected by the decision to adopt one technology rather than another.

Finally, although the technologies we have discussed relate to decision-making in sport, we want to emphasise that they touch on other, more serious, issues about the use of evidence in policy debate. At the time of the first Gulf War the conclusion of a publicised debate over the accuracy of the Patriot anti-missile system, which in part stemmed from the difficulty of interpreting dramatic television footage of missiles being launched into a night sky, had consequences for the credibility of all star-wars-type defence systems (Collins and Pinch, 1998: ch. 1). Since then, debates about video evidence for weapons of mass destruction, the use of computer simulations that predict climate change and the air-brushing of digital images have all highlighted the importance of distinguishing between reality and its digitised representation. With increased computer power and speed these problems will only increase and, before long, it seems likely that the ordinary television viewer will find it almost impossible to tell the difference between synthetic reality and photography. There is, therefore, an urgent need to promote public understanding of these technologies and sports broadcasting, by virtue of its very ubiquity, is an ideal place to start.

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Notes

1. By way of illustration, a search of titles and abstracts for the journal *Public Understanding of Science* carried out on 1 November 2010 returned the following “hits”: sport (1), film (19), television (26), news (64), media (146). The single sport paper is our own paper on Hawk-Eye (Collins and Evans, 2008).
2. It should be noted, however, that the nature, size and significance of the CSI-effect remains the subject of some debate. Ley et al. (2010) provides a review of the relevant literature.
3. There may be other examples of false transparency (e.g. organised cheating) but these do not concern us in this paper.
4. For the notion of “ubiquitous skills” see Collins and Evans (2007).
5. <http://www.guardian.co.uk/sport/2010/jan/04/england-south-africa-third-test-live> (accessed 20 October 2010, emphasis in original). The decision is reported in similar ways on the BBC news website and in the South African newspaper, *The Times*.
6. The full definition is online at: <http://www.lords.org/laws-and-spirit/laws-of-cricket/laws/law-24-no-ball,50,AR.html> (accessed 20 October 2010). The relevant section is paragraph 5(ii).
7. <http://news.bbc.co.uk/sport1/hi/cricket/8452161.stm> (accessed 20 October 2010). Interestingly, the BBC correspondent appears to interpret the rule incorrectly as there is no requirement that the heel be grounded. All that matters is that the foot is behind the line. This highlights the point made earlier that, although the ability to interpret the image requires only ubiquitous skills, a sound knowledge of the relevant rules is also necessary.
8. <http://www.guardian.co.uk/sport/2010/jan/14/south-africa-england-fourth-test-live> (accessed 20 October 2010). The web page also includes two screen captures from Sky Sports that show the relevant delivery.

9. Like the experimenter's regress, the decision would be easy if the answer known but this is precisely what is at stake.
10. <http://www.hawkeyeinnovations.co.uk/UserFiles/File/Hawk-Eye%20accuracy%20and%20believability2.pdf> (accessed 1 November 2010). The quoted material is on page 3; our stress.
11. It is important to stress that this is our only concern. We recognise that there are many other uses of the technology that are of great interest to broadcasters, spectators and coaches and which do not depend on these claims to millimetre accuracy. Examples of these include the graphics produced that show the distribution of scoring shots for particular players (the "wagon wheel") or the tactics used by different bowlers (the "beehive"). All these enhance the viewing experience and contribute to the spectacle and enjoyment of the game.
12. http://www.hawkeyeinnovations.co.uk/?page_id=1011 (accessed 1 November 2010).
13. <http://www.hawkeyeinnovations.co.uk/UserFiles/File/Hawk-Eye%20accuracy%20and%20believability2.pdf> (accessed 1 November 2010).
14. It is likely in any real RTD's distribution of errors that there would be fewer large errors and more small ones. The only information we have on the error distributions of RTDs is that Hawk-Eye's errors do not follow the "normal distribution." We would like to see error distributions published. Here is an error distribution made available to us in respect of TEL (Tennis Electronic Lines), which works from the electrical signals produced by the interaction of a metallised ball and wires buried in the court and about which we have learned much from Henk Jonkhoff. These are the kind of data that all such devices should make available along with notes concerning the exact conditions under which they were generated. In the diagram the horizontal axis shows number of bounces while the vertical axis shows deviation from true bounce point as defined by the mark made on alcohol-moistened talc-covered Perspex.

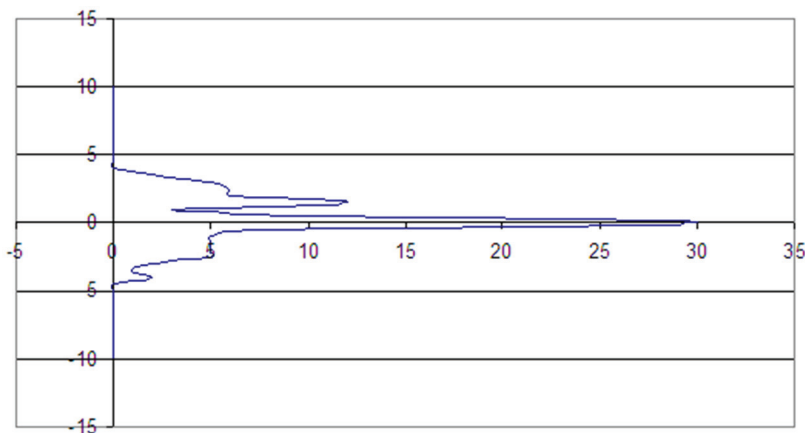


Image reproduced with permission of Henk Jonkhoff.

15. For example, David Gower, ex-captain of the English cricket team, and now a television commentator and newspaper columnist, writes that, in contrast to cricket, Hawk-Eye when used in tennis showed "fact in graphic form" (Gower, 2009).
16. We use the frame rate of 100 fps as this corresponds to the figure given in a recent Hawk-Eye publication, which quotes a figure of 106 fps. <http://www.hawkeyeinnovations.co.uk/UserFiles/File/Hawk-Eye%20accuracy%20and%20believability2.pdf> (accessed 1 November 2010).
17. See Collins and Evans (2008) for a fuller explanation of the lbw rule. The BBC website also has a useful explanation that sets out the problem from the umpire's perspective (http://news.bbc.co.uk/sport1/hi/cricket/rules_and_equipment/6125026.stm) whilst the MCC website has a short guide that explains how the decision review system applies to the lbw decision (<http://www.lords.org/data/files/drs-guide-10354.pdf>). (Both websites accessed 1 November 2010.)

18. Readers of Collins and Evans (2008) will find that this is very close to what we suggested there though we illustrated the point with a shaded rectangle inside the zone of uncertainty.
19. The ball can be up to 22.9 cm in circumference = $c.7.3$ cm in diameter. A stump may be up to 3.81 cm in width.
20. <http://www.lords.org/data/files/drs-guide-10354.pdf> (accessed 1 November 2010).
21. Collins (though not the commentators or anyone else that he has heard from) remains puzzled by one judgement in the South Africa vs. England test again concerning De Villiers' first innings. At one point an appeal for lbw was answered by an "out" call from the on-field umpire. De Villiers asked for a review. Hawk-Eye showed the ball passing over the top of the wicket but also showed that the ball had hit the pad more than 2.5 metres from the wicket. In this case Collins was expecting the on-field umpire's decision to be upheld given that Hawk-Eye had to make an inference over such a long distance and in view of the rule set out above. But the game passed on without remark. Once more, to satisfy Collins that justice had been done he would have had to have heard the conversation between the third umpire and the on-field umpire. (It could well have been that the on-field umpire changed his mind but was he apprised of the fact that the decision had been largely passed to him in view of the distances involved?)
22. Russell (1997) argues that where a big mistake is made a decision is not really a decision at all.
23. As noted earlier, how spectators and viewers interpret these graphics would require a different research project to the one presented here. There is some evidence that some spectators are aware of the issues and we have seen online fan forums and comment boards make reference to the statistical uncertainties associated with Hawk-Eye and similar technologies. On the other hand, Collins was astonished to discover that even senior physicists had not realised they were watching simulations based on inference when they saw the graphic displays produce by an RTD.
24. See e.g. the response to Collins and Evans (2008) by Hawk-Eye Innovations.
25. This solution is a modified version of that put forward in our 2008 paper. There we called for error bars (or error ovals) to be attached to the reconstructed images of the ball. We argued that these error marks should vary according to the speed and direction of the ball, the frame-speed of the cameras, the ambient conditions, and so forth. We no longer believe this is likely to happen. What we have suggested here, however, seems to be easy, quick and cheap to introduce and unobjectionable since it does not require that anyone accept that it proves that RTDs are not accurate, they have only to accept that, as in the case of cricket, ontological authority should remain with the umpire on the case of close calls.
26. There might be a case for making the zone asymmetrical to mimic the tendency of human umpires to see a ball as "out" when it touches the line but skids onwards. As we argue in Collins and Evans (2008), technological aids should change the game as little as possible.
27. As reported on the Gulf News website of 3 March 2007 in respect of a match in Dubai between Nadal and Youzhny. See: <http://gulfnews.com/sport/tennis/hawk-eye-leaves-nadal-and-federer-at-wits-end-1.165119> (accessed 9 February 2010).
28. For a more detailed explanation see Collins (2010).

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